Moiz Asif
SP23-BAI-026
IS Assignment #4

Question 1:-

One round of DES Encryption

Given plaintext (64-bit)

Plain text

0110100001101001011011000110111101101111010111010001101100011101100

given key

0001001100110100010101110111100111001101111011110011011111111110001

Step 1:-
using the given IP table:

Plain text

58	50	42	34	26	18	10	2
60	52	44	36	28	20	12	4
62	54	46	38	30	22	14	6
64	56	48	40	32	24	16	8
57	49	41	33	25	17	9	1
59	51	43	35	27	19	11	3
61	53	45	37	29	21	13	5
63	55	47	39	31	23	15	7

Result after IP

Plain text

1111111100000000111111011010010010000000001111111101111010101010000

CamScanner

Step 2

split into L_0 and R_0

Left half L_0 (first 32 bites)

Plain text

11111111000000000111111101101001 0

Right half R_0 (last 32 bites)

Plain text

00000000111111111011110101010000

Step 3:-

Expansion permutation (EIP) on R_0

Given EIP Table:-

Plain text:-

32	1	2	3	4	5
4	5	6	7	8	9
8	9	10	11	12	13
12	13	14	15	16	17
16	17	18	19	20	21
20	21	22	23	24	25
24	25	26	27	28	29
28	29	30	31	32	1

R_0

Plain text:-

00000000111111101111010101010000

CamScanner

After expansion:

Expanded R_0 (48 bits)

Plain text

000000000001011111111111101111111 010101010100000

Step 4:-

generate first subkey k_1

Using the DES key scheduling process, first subkey generated from the given key

k_1

Plain text

000110110000001011101111111100011100000111010

Step 5:-

XOR expanded R_0 with k_1

XOR operation

Plain text

000000000000101111111111111011111 010101010100000

000110110000000010111011111111 000111000001110010

000110110001010100010000001 000111101101011010010

Final result After one DES round step

XOR output

Plain text

00011011000101010001000000100011110111010110110010

Question #2

DES key scheduling

given hexadecimal key

Plain text

0F1571C947D9E859

convert hexadecimal to Binary

Hex → Binary.

0 = 0000
F = 1111
1 = 0001
5 = 0101
7 = 0111
1 = 0001
C = 1100
9 = 1001
4 = 0100
7 = 0111
D = 1101
9 = 1001
E = 1110
8 = 1000

5 = 0101
9 = 1001

CamScanner

Complete 64-bit binary key

0000111100010101011100011100101001000111111011001111
0100001011001

Apply PC1:-

After applying PC1

C_0

0111010001111110001000100101010

D_0

1000010001011111101001011000

left shift by 1 bit

C_1

1101000011111100010001001010100

D_1

0000100010011111101001011000

Apply PC_2

Combine C_1 and D_1.

1101000011111100010001001010000001000100111110100
10110001

First subkey k1

final result:-

011110000011001111000011001000000110 110100
1110000

CamScanner

Question #3

AES operations

Plain text:-

32	43	F_6	A8
88	5A	30	8D
31	31	98	A2
E_0	37	07	34

Round key:-

2B	7E	15	16
28	AE	D2	A6
AB	F7	15	88
09	CF	4F	3C

Arrange plaintext into AES state matrix

AES uses column-major order.

State Matrix

$$\begin{bmatrix} 32 & 88 & 31 & E_0 \\ 43 & 5A & 31 & 37 \\ F6 & 30 & 98 & 07 \\ A8 & 8D & A2 & 34 \end{bmatrix}$$

Arrange round key into key matrix

key matrix:

CamScanner

$$\begin{bmatrix} 2B & 28 & AB & 09 \\ 7E & AE & F7 & CF \\ 15 & D2 & 15 & 4F \\ 16 & A6 & 88 & 3C \end{bmatrix}$$

Add round key operation

complete Addroundkey result

$$\begin{bmatrix} 19 & A0 & 9A & E9 \\ 3D & F4 & C6 & F8 \\ E3 & E2 & 8D & 48 \\ BE & 2B & 2A & 08 \end{bmatrix}$$

Shift rows operation

After shift rows

$$\begin{bmatrix} 19 & A0 & 9A & E9 \\ F4 & C6 & F8 & 3D \\ 8D & 48 & E3 & E2 \\ 08 & BE & 2B & 2A \end{bmatrix}$$

Mix columns – First byte of first column

first column

$$\begin{bmatrix} 19 \\ FA \\ 8D \\ 08 \end{bmatrix}$$

CamScanner

AES formula for first byte

$(02 \times 19) \oplus (03 \times F4) \oplus (01 \times 80) \oplus (01 \times 08)$

Galois field Multiplication

1. 02 x 19

$19 = 00011001$

shift left

$00110010 = 32$

So,

$02 \times 19 = 32$

2. 03 x F4

$03 \times x = (02 \times x) \oplus x$

$F_4 = 11110100$

left shift $\rightarrow$ 11101000

MSB was 1 so XOR with 1B

$EB \oplus 1B = F_3$

So,

$02 \times F_4 = F_3$

Now

$F_3 \oplus F_4 = 07$

So,

$03 \times F_4 = 07$

CamScanner

Remaining terms

$01 \times 8D = 8D$

$01 \times 08 = 08$

XOR All results

$32 \oplus 07 = 35$

$35 \oplus 8D = B8$

$B8 \oplus 08 = B0$

(f) Why AES uses Multiple Transformations

AES uses several transformations because a single XOR operation is not secure enough.

Purpose of each Transformation

1. Add Round Key

- Introduces the secret key into encryption
- Makes ciphertext dependent on the key.

2. Shift rows

- Rearranges bytes
- create diffusion by spreading data across columns.

3. Mix Columns

CamScanner

- Mixes bytes mathematically using galois feild arithmetic
- Create strong diffusion and interdependency

4. Multiple rounds

- Prevents attackers from discovering patterns.
- provide confusion and diffusion together.

CamScanner